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Mahindra University, Hyderabad

Ecole Centrale School of Engineering

Project Report

"Insight Flow SQL Analytics Business Intelligence & AI-Powered Analytics Platform"

Program: M.Tech

Branch: AI & DS

Subject: Gen AI

Semester: II

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Insight Flow SQL Analytics
Business Intelligence & AI-Powered Analytics Platform

1. Introduction

The rapid growth of e-commerce platforms has generated massive amounts of customer transaction data. Businesses require intelligent analytics systems to transform raw transactional records into meaningful insights for decision-making.

This project, **Insight Flow SQL Analytics**, is an AI-powered business intelligence dashboard developed using:

- Python
- SQL (SQLite)
- Streamlit
- Plotly
- Machine Learning
- Pandas
- Scikit-Learn

The system performs:

- Data Cleaning
- SQL Data Pipeline Creation
- Feature Engineering
- Business Analytics
- RFM Customer Analysis
- Revenue Forecasting
- Customer Segmentation
- Interactive Dashboard Visualization

The project provides a real-time analytics environment where users can analyze sales, customers, revenue trends, and AI-generated insights through an interactive dashboard.

2. Problem Statement

Traditional business reporting systems provide static reports that lack:

- Interactivity
- Real-time analytics
- Intelligent forecasting
- Customer segmentation
- AI-driven insights

The objective of this project is to develop an intelligent analytics platform capable of:

- Processing raw e-commerce datasets
- Creating analytical SQL pipelines
- Generating visual insights
- Performing predictive analytics
- Supporting business decision-making

3. Objectives

The main objectives of the project are:

- To clean and preprocess e-commerce datasets
- To create SQL-based analytical pipelines
- To build an interactive BI dashboard
- To visualize business metrics dynamically
- To perform RFM customer analysis
- To implement machine learning models for:
  - Revenue Prediction
  - Customer Segmentation
- To provide AI-based business insights

4. Dataset Description

The project uses the Brazilian E-Commerce Public Dataset (Olist Dataset).

Datasets Used:

Dataset	Description
orders	Order information
customers	Customer details
order_items	Product purchase details
payments	Payment information
products	Product details

Total Records:

Table	Rows
Orders	99,441
Customers	99,441

Table	Rows
Order Items	112,650
Payments	103,886
Products	32,951

5. Technologies Used

Technology	Purpose
Python	Core Programming
Pandas	Data Processing
SQLite	Database Management
Streamlit	Dashboard Development
Plotly	Data Visualization
Scikit-Learn	Machine Learning
NumPy	Numerical Computation
Google Colab	Development Environment
Ngrok	Public URL Tunneling
GitHub	Version Control

6. System Architecture

Workflow:

1. Data Loading
2. Data Cleaning
3. SQL Table Creation
4. Feature Engineering
5. Data Merging
6. Analytics Processing
7. Dashboard Visualization
8. Machine Learning Analytics
9. AI Insights Generation

7. Data Cleaning

The raw datasets contained:

- Missing values
- Duplicate records
- Date format inconsistencies

Cleaning Operations Performed:

- Converted timestamps into datetime format
- Removed duplicate rows
- Removed rows with critical missing delivery dates
- Standardized date columns

Before Cleaning:

- Missing values existed in:
    - order_approved_at
    - order_delivered_carrier_date
    - order_delivered_customer_date

After Cleaning:

The missing values were significantly reduced and the dataset became suitable for analytics and ML processing.

8. SQL Data Pipeline

SQLite was used as the backend analytical engine.

SQL Tables Created:

- orders
- customers
- order_items
- payments
- products

SQL Operations Used:

- SELECT
- JOIN
- GROUP BY

- ORDER BY
- Aggregations
- CASE statements
- Date functions

9. Feature Engineering

Feature engineering was performed to derive analytical attributes.

Features Created:

1. Total Payment Aggregation: SUM(payment_value)

2. Delivery Delay:

julianday(order_delivered_customer_date) **-** julianday(order_estimated_delivery_date)

3. RFM Metrics:

- Recency
- Frequency
- Monetary

10. Data Merging

A central analytical table named: fact_orders, was created using SQL JOIN operations.

Tables Merged

- Orders
- Customers
- Order Items
- Payments

This enabled efficient analytics querying.

11. Dashboard Development

The dashboard was developed using Streamlit.

Dashboard Features:

- **Sidebar Filters**
  - State Filter
  - Date Range Filter

- **KPI Cards**
  - Total Revenue
  - Total Orders

- Total Customers
- Average Order Value
- Repeat Customer Percentage
- Average Daily Orders

12. Data Visualizations

The dashboard contains multiple interactive visualizations.

Charts Implemented:

1. Orders Over Time

- Line chart showing order trends.

2. Revenue by State

- Bar chart showing top revenue-generating states.

3. Revenue Distribution

- Pie chart showing revenue contribution.

4. Monthly Revenue Trend

- Time-series revenue analysis.

5. Orders by Weekday

- Weekday-based order distribution.

6. Top Cities by Revenue

- Horizontal bar chart.

7. Orders Heatmap

- Month vs Weekday heatmap.

8. Customer Segmentation Scatter Plot

- Cluster-based customer segmentation.

13. RFM Analysis

RFM Analysis was used for customer behavior analysis.

RFM Components:

- **Recency:** How recently the customer purchased.
- **Frequency:** How often the customer purchases.
- **Monetary:** How much money the customer spends.

RFM Scoring:

Customers were scored using quartile-based ranking.

- R_score

- F_score
- M_score

These scores were combined into: RFM_Score

14. Talk-To-AI Module

A simple NLP-based SQL query assistant was implemented.

Users can ask questions like:

- Total revenue
- Top customers
- Top cities

The system dynamically converts queries into SQL commands.

15. AI Insights Module

The system automatically generates intelligent business insights such as:

- Highest revenue-generating state
- Best-performing weekday
- Top-performing city
- Business recommendations

Example:

- Friday has peak customer activity.
- Increase ad campaigns and discounts on Fridays.

16. Machine Learning Implementation

The project includes two machine learning modules.

- **Revenue Forecasting**

Algorithm Used: Linear Regression

The model predicts future revenue trends based on historical monthly revenue.

Steps

1. Aggregate monthly revenue
2. Convert months into numerical values
3. Train Linear Regression model
4. Predict future revenue

Output

- Actual Revenue Trend
- Predicted Revenue Trend

Benefits

- Sales forecasting

- Budget planning
- Business growth estimation

➢ **Customer Segmentation**

Algorithm Used: K-Means Clustering

Customers were grouped into clusters based on:

- Frequency
- Monetary value

Segments Created:

- Regular Customers
- Premium Customers
- High Value Customers

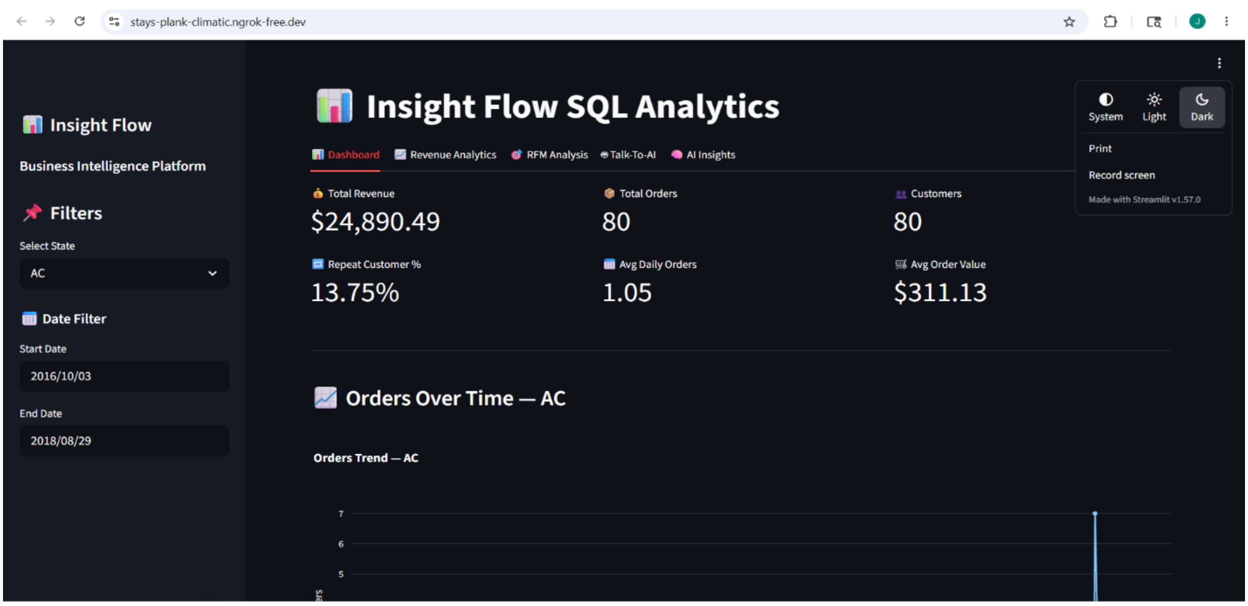
Benefits:

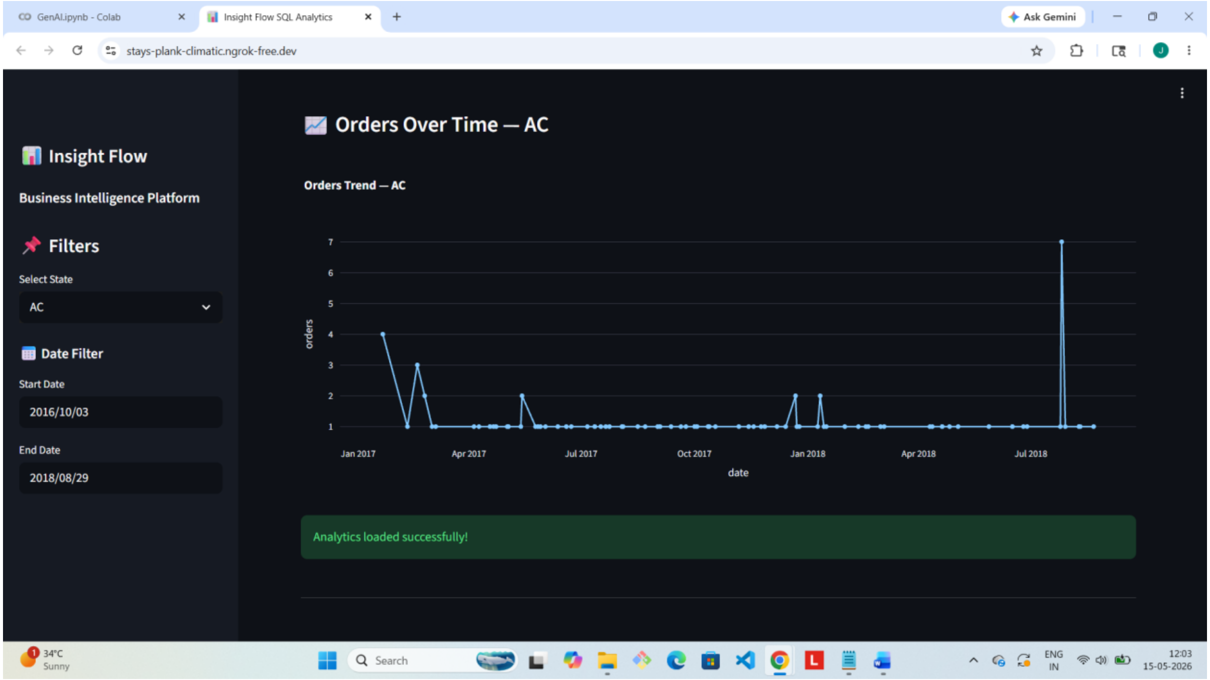
- Personalized marketing
- Customer targeting
- Loyalty analysis

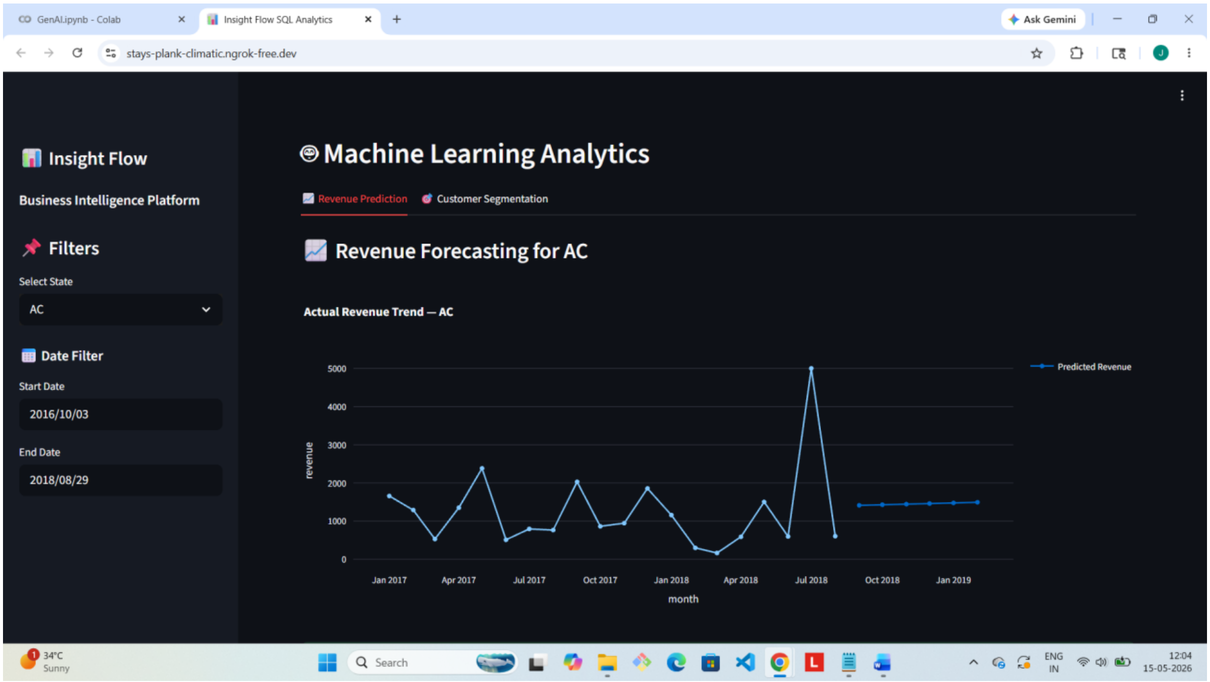
17. UI/UX Features

The dashboard includes:

- Dark Theme UI
- Custom CSS Styling
- Interactive Plotly Charts
- Responsive Layout
- Animated Loading Spinner
- Styled Metric Cards
- Green Glow Input Effects
- Professional Sidebar Navigation







18. Backend Features

The backend supports:

- SQL Query Processing
- Dynamic Filtering
- Real-Time Aggregations
- Data Pipeline Management
- ML Integration
- Automated Insights Generation

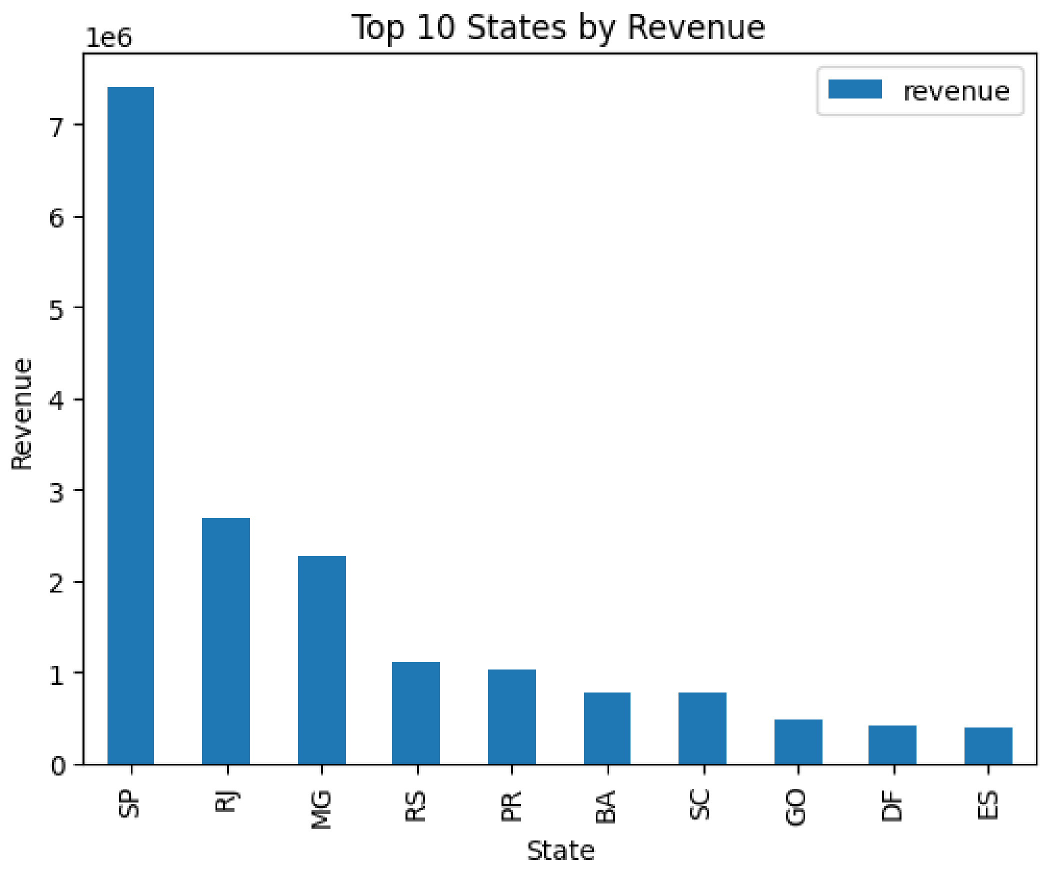
19. Results

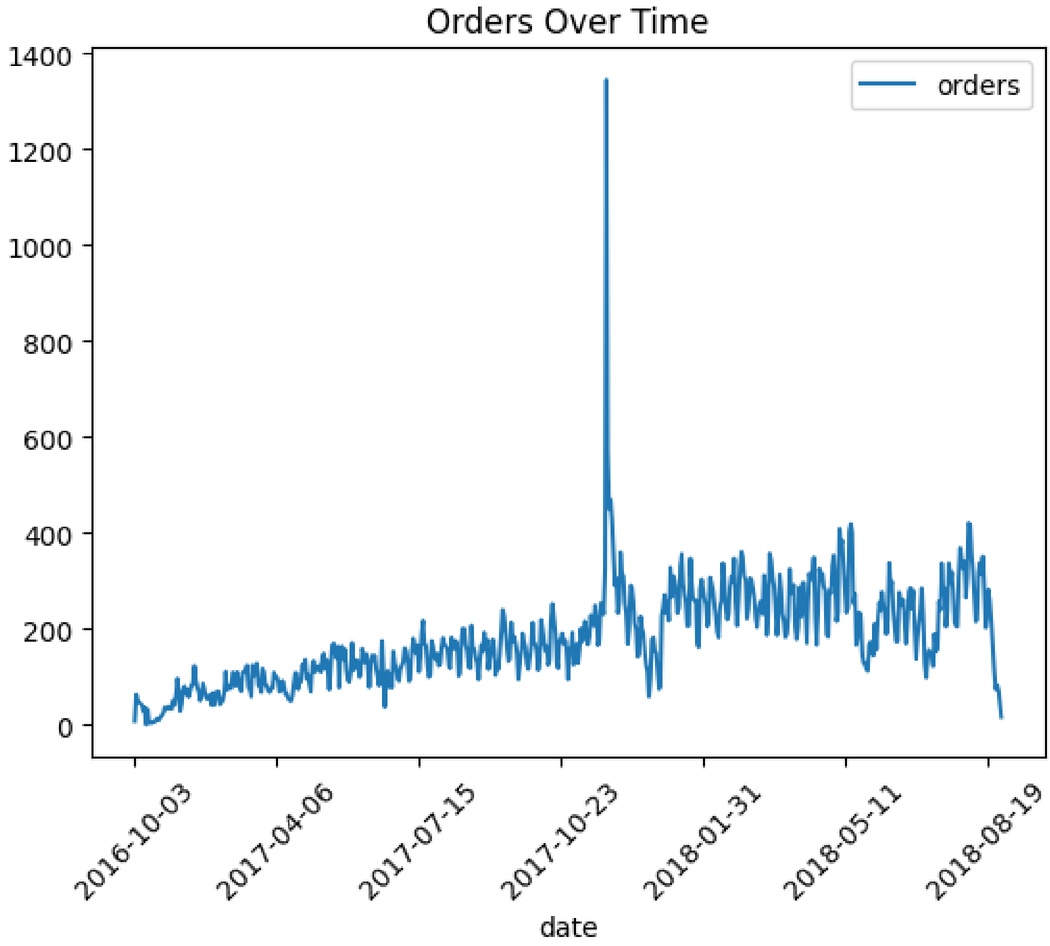
The project successfully achieved:

- End-to-end analytics pipeline
- Interactive BI dashboard
- SQL-based analytics engine
- Machine learning forecasting
- Customer segmentation
- AI-generated business insights

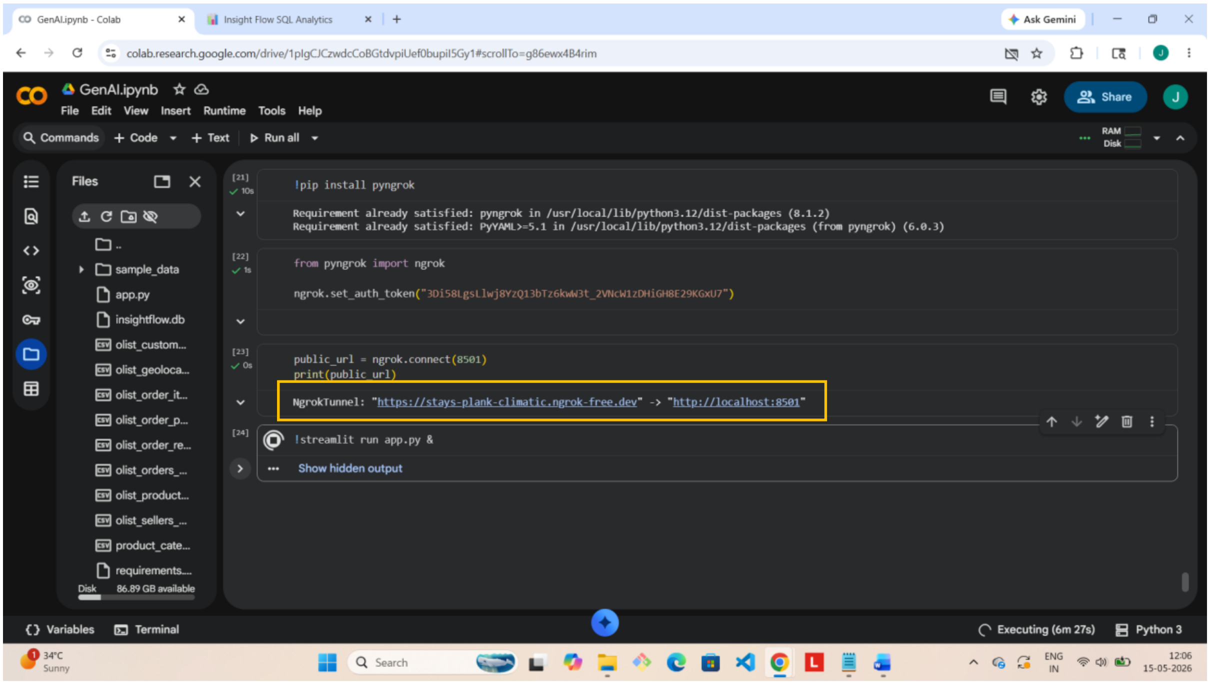
The dashboard enables business users to:

- Monitor revenue
- Analyze customer behavior
- Predict future sales
- Make strategic business decisions





Orders Over Time
1400
1200
1000
800
600
400
200
0
orders
2016-10-03
2017-04-06
2017-07-15
2017-10-23
2018-01-31
2018-05-11
2018-08-19
date



GenAI.ipynb - Colab
Insight Flow SQL Analytics
Ask Gemini
colab.research.google.com/drive/1plgCJCzwdcCoBGtdvpiUef0bupil5Gy1#scrollTo=g86ewx4B4rim
GenAI.ipynb
File Edit View Insert Runtime Tools Help
Share
Commands
+ Code
+ Text
Run all
RAM
Disk
Files
sample_data
app.py
insightflow.db
olist_custom...
olist_geoloca...
olist_order_it...
olist_order_p...
olist_order_re...
olist_orders_...
olist_product...
olist_sellers_...
product_cate...
requirements...
Disk 86.89 GB available
[21]
!pip install pyngrok
Requirement already satisfied: pyngrok in /usr/local/lib/python3.12/dist-packages (8.1.2)
Requirement already satisfied: PyYAML>=5.1 in /usr/local/lib/python3.12/dist-packages (from pyngrok) (6.0.3)
[22]
from pyngrok import ngrok
ngrok.set_auth_token("3Di58LgsLlwj8YzQ13bTz6kwW3t_2VNcW1zDHiGH8E29KGxU7")
[23]
public_url = ngrok.connect(8501)
print(public_url)
NgrokTunnel: "https://stays-plank-climatic.ngrok-free.dev" -> "http://localhost:8501"
[24]
!streamlit run app.py &
Show hidden output
Variables
Terminal
Executing (6m 27s)
Python 3
34°C
Sunny
Search
ENG
IN
12:06
15-05-2026

20. Advantages

- Real-time analytics
- Interactive visualizations
- AI-driven recommendations
- Intelligent customer segmentation
- Revenue forecasting
- Easy-to-use dashboard
- Scalable architecture

21. Limitations

- Talk-To-AI module supports limited query types
- SQLite is suitable for small-to-medium scale data
- Revenue prediction uses basic linear regression
- No cloud deployment implemented

22. Future Enhancements

Future improvements may include:

- GPT-powered natural language querying
- Advanced ML models
- Deep learning forecasting
- Cloud deployment
- User authentication
- Real-time streaming analytics
- Power BI integration
- Recommendation systems

23. Conclusion

Insight Flow SQL Analytics successfully demonstrates the integration of:

- Data Engineering
- SQL Analytics
- Business Intelligence
- Machine Learning
- Interactive Visualization

The system transforms raw e-commerce data into meaningful business insights through a modern analytics dashboard.

This project highlights practical implementation of:

- SQL pipelines
- Data analytics
- RFM analysis
- Revenue prediction
- Customer segmentation
- AI-driven insights

The platform can be extended into a production-grade business intelligence solution for enterprise analytics.

Github Link : https://github.com/Jessicadollz/Insight-Flow-SQL-Analytics.git